



Tool Documentation

INDUSTRIAL ENERGY EFFICIENCY AND PRODUCTIVITY TOOLKIT

USER GUIDE FOR KMM TOOL

September 2026

Authored by: Lawrence Berkeley National Laboratory and University of California, Davis

Contact:

Prakash Rao, prao@lbl.gov

J. Kelly Kissock, jkissock@ucdavis.edu

Table of Contents

Introduction	2
How to Use the Tool.....	3
Step 1a Custom Hourly Input Template	3
Step 1b Prepare the hourly input sheet	3
Step 2 Green Button input	4
Upload Data and Select Clusters	4
Step 3 Upload the dataset.....	4
Step 4 Review and adjust the number of clusters.....	4
Output.....	5
Cluster Profiles	5
Cluster Occurrences Heatmap	6
Box and Whisker Plots.....	7
Cluster Details Data.....	8
Worked Example	9
Prepare the example and run the analysis.....	9
Interpret and Refine Example	10
Compare the representative profiles	10
Check the calendar and variability	10
Compare different cluster counts.....	10

Introduction

The K-Means Clustering Tool (KMM) groups days (24-hour periods) with similar hourly profiles so that a facility can review a set of representative operating patterns. Each day is described by statistical characteristics of its hourly values, such as electric demand (kW). KMM can also be used with thermal use, water use, production, or another quantity recorded hourly. The tool aids in identifying common operating patterns, comparing variability, and selecting days or periods for further investigation. Use operating schedules and meter records to determine what the patterns represent.

K-means is an unsupervised clustering method that divides observations into a specified number of groups called clusters. In the tool, each day's 24-hourly readings form a daily profile, and days with similar profiles are grouped together. The tool uses the original values without normalizing each day, so both the magnitude and the shape of the profile influence clustering. For example, two days might have the same pattern (low consumption overnight and a peak at noon) but one uses twice as much electricity throughout the day. KMM can place them in different clusters because their magnitudes differ, even though their shapes are similar. When sufficient data are available, KMM evaluates two through ten clusters and selects the number with the highest average silhouette score. The silhouette score measures how well each day fits within its assigned cluster compared with the nearest other cluster. Scores range from -1 to +1: values near +1 indicate a good fit, values near 0 indicate overlap between clusters, and negative values suggest the day may fit better in another cluster. KMM averages these scores across days and uses the highest average to select the suggested number of clusters. Users can also adjust the number of clusters manually after loading the data (see Figure 1).

This guide explains how to prepare and upload data, select the number of clusters, interpret the outputs, and download results. The worked example uses the template's example values; replace them with facility data for an assessment.

How to Use the Tool

Follow the steps below in order. In Figure 1, the left sidebar contains input type, its date period, input excel sheet download, and an excel sheet upload button. Other instructions regarding the tool are also shown in the “Dashboard” tab.

K-Means Clustering Tool

1. Unit & Format Configuration

Unit of Analysis (e.g. kW, MMBtu, tons, or gallons):

Select Data Format:

☒ Custom Hourly Input Template

☐ Green Button Data (.XML)

Select Template Boundaries:

to

[Download Custom Hourly Input Sheet](#)

2. Upload Dataset

Choose File:

No file selected

Hourly Profile Clustering Tool

Extract representative daily profiles from long-term data for your facility (For example: interval meter logs, thermal profiles, water usage, or production outputs etc.) using K-Means clustering.

Getting Started:

- Specify your custom unit of analysis in the sidebar panel (e.g. kW, MMBtu, tons, or gallons).
- Specify start and end date of your dataset to export an input sheet template or upload Green Button XML data as your input.
- Upload your file using the Browse button in the sidebar panel.
- The tool will auto-compute the optimal number of clusters for the uploaded profile based on the Silhouette method to establish the baseline cluster profile count.
- Modify the number of clusters if the optimum Silhouette clusters do not fully capture the variability in your uploaded profile (optional).

BERKELEY LAB
Prakash Rao
prao@lbl.gov

UC DAVIS
Energy and Efficiency Institute
Kelly Kissock
jkissock@ucdavis.edu

Figure 1: KMM Landing Page

Step 1a Custom Hourly Input Template

1. Enter the Unit of Analysis, such as kW, MMBtu, tons, or gallons. Use a unit that describes the numeric values in your file.
2. Under Select Data Format, choose Custom Hourly Input Template for an Excel workbook or Green Button Data (.XML) for compatible meter data.

The unit field changes chart labels, hover labels, and export headings only. It does not convert the values, normalize the profiles, or change the calculation. For example, changing “kW” to “MW” does not divide the readings by 1,000.

Step 1b Prepare the hourly input sheet

1. Select Custom Hourly Input Template. Set the start and end dates under Select Template Boundaries.
2. Click Download Custom Hourly Input Sheet. Open Custom_Hourly_Input_Sheet.xlsx and use the Data Input worksheet.

3. Keep the first column, Date_Time, aligned with your timestamps. Replace all example numbers in the yellow second column with your input.
4. Save the completed workbook.

Table 1: Example layout from the downloaded template. These numbers are demonstration values, not measurements for your facility.

Date_Time	Input kW
2026-01-01 00:00:00	276.51
2026-01-01 01:00:00	273.67
2026-01-01 02:00:00	276.57

Step 2 Green Button input

Green Button is a standardized format that allows utility customers to download their electricity, natural gas, or water usage data and use it in tools to better understand their consumption patterns.¹

To use this option in KMM, log in to your utility's online account portal and download hourly Green Button data in XML format for the meter and date range you want to analyze. In KMM, select Green Button Data (.XML), click "Browse" under Choose File, and upload the downloaded file. Availability of Green Button downloads depends on your utility, so ensure that the corresponding utility participates in this program.

Upload Data and Select Clusters

Step 3 Upload the dataset

1. Select the data format that matches your prepared file.
2. Under Choose File, click Browse and select the workbook or XML file. Wait for processing to finish
3. Confirm that the Dashboard displays results and that the sidebar shows Modify Clusters (Optional). Review the day counts before interpreting the patterns.

Step 4 Review and adjust the number of clusters

The Number of Clusters (k) input controls how many representative profiles are produced. When there is enough usable data for automatic search, KMM tests k values from 2 through 10 and selects the value with the highest average silhouette score. The sidebar identifies this selection with "Optimum k". The automatic search requires more than 100 valid input rows

¹ <https://www.greenbuttonalliance.org/faqs>

and more than 15 dates in its daily matrix. It can also fail if there are too few distinct daily profiles. If no automatic selection appears, review the data and choose a feasible k manually.

You may enter an integer from 1 through 10. Use fewer clusters to summarize broad patterns and more clusters to investigate additional differences. Keep k within the available number of distinct daily profiles. A higher silhouette score generally indicates better separation. It does not establish the operational meaning of the groups. Review the centroid shapes, spread, day counts, and calendar pattern before deciding which k best supports your assessment. Cluster labels are assigned in order to decrease the day count. Cluster 1 is the most frequent group, not necessarily the highest-load group. Changing k or the input dataset can change both labels and membership.

Output

Cluster Profiles

The Cluster Profiles chart is the first chart on the Dashboard. Each panel represents a cluster, and its heading reports the number of days assigned to that cluster. Hours are shown from 0 through 23 in Figure 2.

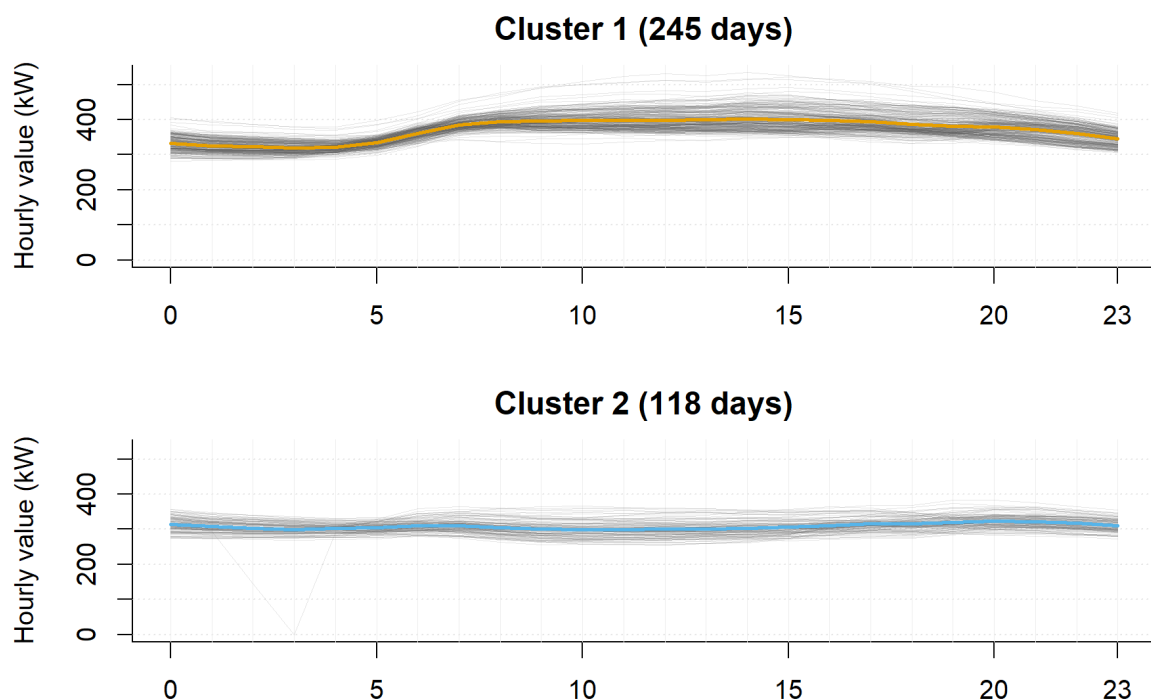


Figure 2: Representative daily profiles for the worked example

- The colored line is the mean value at each hour across the days in the cluster. It is the centroid, or representative profile.
- The gray lines are the individual daily profiles. A wide variation indicates differences that the average line alone may hide.

- Hover over the colored line to read the hour and mean hourly value. Compare the timing and magnitude of peaks and the overnight baseline.

Use these patterns to choose follow-up questions. For example, compare a recurring daytime rise with production schedules or review a sustained overnight load with the operating team. Click [Download View \(PNG\)](#) for the profile figure or [Download Chart Data \(CSV\)](#) for the 24-hour centroid values for each cluster.

Cluster Occurrences Heatmap

The heatmap places the assigned cluster for each retained day into a calendar-style display. The horizontal axis is day of week, the vertical axis is week starting date, and the cell colors correspond to the clusters in the profile plots (Figure 3).

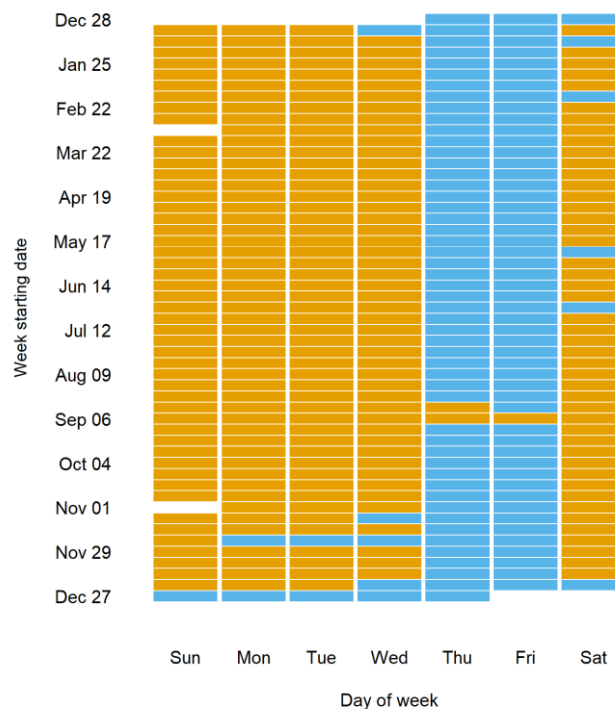


Figure 3: Calendar distribution of clusters for the worked example

Look for repeating weekday and weekend patterns, seasonal changes, isolated days, and extended periods assigned to a different group. Hover over a cell to see its date, assigned cluster, and cluster median. The hover value labelled Cluster Median is the median of daily mean values for that cluster. It matches the Cluster Analysis box plot, rather than the Median field in the engineering metrics table.

Blank positions may be outside the uploaded period or may correspond to days excluded from the analysis. They are not evidence of zero consumption. Compare the displayed day counts with the expected number of complete days. Click [Download Heatmap View \(PNG\)](#) to save the figure. The current interface does not provide a separate date-to-cluster CSV export.

Box and Whisker Plots

The Box and Whisker Plots section contains two tabs (Figure 4). Both show the middle half of the values as a box (Quartile 1-Quartile 3), the median as a line, and the minimum and maximum as the ends of the whiskers. The whiskers use the full observed range, not a 1.5-times-interquartile-range cutoff.

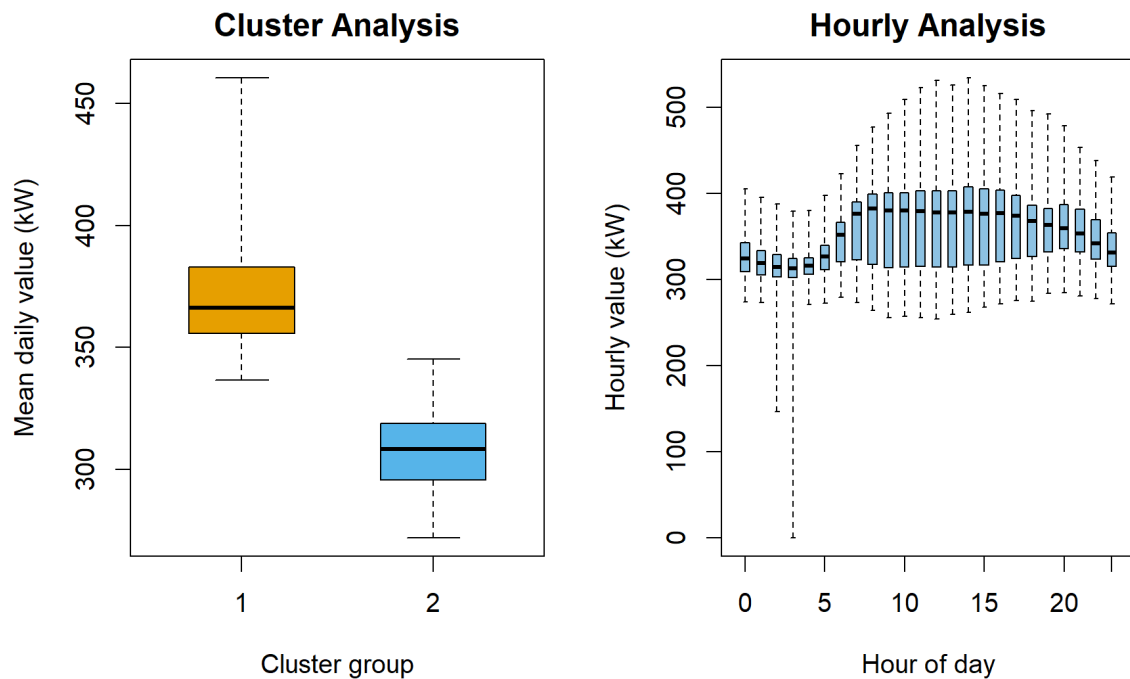


Figure 4: Cluster and hourly distributions for example worked

The first tab (cluster summary) calculates one mean value for each retained day, then summarizes those daily means within each cluster. Use it to compare typical daily operating levels and the spread among days assigned to the same group. Hover to see the minimum, median, and maximum daily mean.

The second tab (hourly summary) summarizes the readings at each hour of the day across all retained days. It combines the clusters. Use it to identify hours with a narrow operating range or substantial variation from day to day. Hover to see the minimum, median, and maximum hourly reading.

These views summarize different populations. The Cluster Analysis median and range are calculated from daily means; the Hourly Analysis values come from readings at a specific hour. The Download Box Plot View (PNG) button exports the Cluster Analysis figure even when the Hourly Analysis tab is selected. The Download Chart Data (CSV) button exports the engineering summary table; it does not export the quartiles shown in either box plot.

Cluster Details Data

Open the Cluster Details Data tab (second major tab at the top of the dashboard) to view the Extracted Engineering Parameters Table. The table contains one row per cluster. Its values describe the retained hourly records.

Table 2: Summary of table parameters in the cluster details data tab

Field	Meaning
Cluster	Group label, ordered by decreasing number of assigned days.
Number of Days	Count of distinct retained dates assigned to the cluster.
Mean	Average of all hourly readings in the cluster.
Median	Median of all hourly readings in the cluster.
Min and Max	Lowest and highest individual hourly readings in the cluster.
Standard Deviation	One sample standard deviation of all hourly readings in the cluster.
Centroid Peak Hour	Hours from 0 to 23 at which the cluster's mean hourly profile reaches its maximum. One hour is returned when peaks tie.

The table's Median, Min, and Max can differ from the same labels in the Cluster Analysis box plot because the table uses hourly readings and the plot uses daily means. The table's Standard Deviation includes variation across hours and across days, not only variation among daily averages.

For hourly electrical demand in kW, multiplying a complete day's mean kW by 24 hours gives that day's energy in kWh. KMM does not add this total energy to the table. For other quantities, interpret the mean and any derived totals according to the physical meaning of the input. Click Download Metrics Log (CSV) in the sidebar to save this table. The same table is also available from Download Chart Data (CSV) below the box plots.

Worked Example

Prepare the example and run the analysis

This example uses the baseline values supplied by the KMM template. It demonstrates the workflow and interpretation; it is not a forecast of facility consumption or a savings estimate.

1. Open the sandbox site. Keep Unit of Analysis as kW and select Custom Hourly Input Template.
2. Set Select Template Boundaries to January 1 through December 31, 2026. Download the custom hourly input sheet.
3. For this demonstration, retain the supplied example values supplied. For facility work, replace every example value with the corresponding measured value.
4. Upload the workbook. Let KMM select the baseline k, then check the profile day counts and the Cluster Details Data table.

For the downloaded example evaluated here, the highest average silhouette score among k = 2 through 10 is approximately 0.5955 at k = 2, thereby selecting “2” clusters (see Table 3).

Table 3: Summary of the retained hourly readings in the worked example

Cluster	Days	Mean kW	Median kW	Peak hour
1	245	371.41	373.39	14
2	118	308.29	307.03	20

The first cluster contains 245 days, approximately 67.5% of the retained days. Its mean is 371.41 kW, and its centroid peak occurs at 2 PM. The second contains 118 days, approximately 32.5%, with a mean of 308.29 kW and a centroid peak at 8 pm in the evening. The expected number of days depends on the timestamps actually present in the downloaded file and how those timestamps are interpreted. Always reconcile the retained-day count with the input. Do not assume that an annual workbook contributes 365 complete days.

Interpret and Refine Example

Compare the representative profiles

In Figure 2, the more frequent cluster operates at a higher average level. Compare its daytime peak and overnight baseline with the second cluster. Review the gray traces to see how well each centroid represents the assigned days.

Check the calendar and variability

Use Figure 3 to identify repeated calendar patterns and dates worth reviewing. Cross-check those dates against shift schedules, shutdowns, weather, production records, and meter quality.

Use Figure 4 to distinguish the spread in daily means from the spread at a particular hour. A cluster may have a narrow distribution of daily means while still having substantial differences within the day.

Compare different cluster counts

1. Record the baseline k , its score, and the number of days in each group.
2. Change Number of Clusters (k) from 2 to 3. Wait for the charts to update, then compare profile shapes, day counts, and the calendar pattern.
3. Decide whether the additional group distinguishes an operationally useful pattern. A smaller group may reveal a useful exception, but it can also reflect unusual or poor-quality data.

For this example, the precomputed average silhouette score at $k = 3$ is approximately 0.5065, lower than the $k = 2$ score. That comparison supports retaining two groups for the baseline summary while leaving three groups available for a more detailed review.